

SCAM REPORT PLATFORM

Wideband Delphi Estimation

1. Estimation Team

Student ID	Name	Code	Team Role
67130500846	Aekarut Phetpradit	A.P.	Fullstack
67130500856	Thanapoom Palachewa	T.P.	Fullstack
67130500866	Shapphanyu Phaniphad	S.P.	Fullstack
67130500841	Benyapon Saisong	B.S.	Fullstack
67130500863	Yossapan Rujipornprasert	Y.R.	Fullstack

2. Round 1 — Independent Estimation

WBS ID	Work Package	A.P.	T.P.	S.P.	B.S.	Y.R.	Mean
1.1.1	Authentication	5	4	6	5	5	5.0
1.1.2	Scam Report Feed	7	5	6	6	6	6.0
1.1.3	AI-Powered Search UI	13	15	14	10	14	13.2
1.1.4	Report Submission Form	7	10	8	9	5	7.8
1.1.5	Announcements Feature	7	8	8	5	5	6.6
1.1.6	Data Privacy & ToS	5	6	4	3	5	4.6

WBS ID	Work Package	A.P.	T.P.	S.P.	B.S.	Y.R.	Mean
1.1.7	Support & Settings	3	5	4	8	4	4.8
1.2.1	Pending Review Queue	7	6	6	6	6	6.2
1.2.2	Admin Action UI	7	7	8	8	3	6.6
2.1.1	PostgreSQL + pgvector	7	7	7	7	7	7.0
2.1.2	Prisma ORM Implementation	5	5	6	5	3	4.8
2.2.1	Firebase Auth Integration	5	5	5	4	5	4.8
2.2.2	Supabase Storage Integration	5	5	5	7	5	5.4
2.3.1	Gemini LLM Integration	5	7	5	4	6	5.4
2.3.2	RAG Retrieval Logic	11	12	12	11	12	11.6
2.3.3	API Services	13	12	13	12	12	12.4
3.1	Unit & Integration Testing	7	6	7	6	7	6.6
3.2	User Acceptance Testing	5	5	5	5	5	5.0
3.3	CI/CD Pipeline & Deployment	5	5	5	5	5	5.0
	TOTAL	129	135	133	126	120	128.6

3. Discussion Notes

We have taken the time to go through each feature in detail so that everyone can clearly understand how they work and share the same overall view of the application's flow and design. This discussion helped ensure that all team members are aligned and have a consistent understanding of how the system should function.

We also agreed that, for the next round of planning and estimation, we will use average hours as a basis for our calculations. This approach should make our estimates more realistic and fair, taking into account the complexity of each feature as well as the varying skill levels within the team.

4. Round 2 — Revised Estimation

WBS ID	Work Package	A.P.	T.P.	S.P.	B.S.	Y.R.	Mean
1.1.1	Authentication	5	5	5	5	5	5.0
1.1.2	Scam Report Feed	6	7	6	7	6	6.4
1.1.3	AI-Powered Search UI	11	12	12	12	12	11.8
1.1.4	Report Submission Form	7	9	8	8	8	8.0
1.1.5	Announcements Feature	7	7	7	6	7	6.8
1.1.6	Data Privacy & ToS	5	4	4	3	4	4.0
1.1.7	Support & Settings	5	4	4	5	4	4.4
1.2.1	Pending Review Queue	7	6	7	8	7	6.8
1.2.2	Admin Action UI	7	7	7	8	7	7.2
2.1.1	PostgreSQL + pgvector	7	7	7	6	7	6.8
2.1.2	Prisma ORM Implementation	5	5	6	6	6	5.6
2.2.1	Firebase Auth Integration	5	5	5	10	6	6.2
2.2.2	Supabase Storage Integration	5	5	6	12	6	6.8
2.3.1	Gemini LLM Integration	7	7	7	8	7	7.2
2.3.2	RAG Retrieval Logic	13	12	12	7	12	11.2
2.3.3	API Services	11	12	12	11	12	11.6
3.1	Unit & Integration Testing	5	6	6	6	5	5.6

WBS ID	Work Package	A.P.	T.P.	S.P.	B.S.	Y.R.	Mean
3.2	User Acceptance Testing	5	5	5	6	5	5.2
3.3	CI/CD Pipeline & Deployment	5	5	5	6	5	5.2
	TOTAL	128	130	131	140	131	132.2

5. Final Consensus Estimates

WBS ID	Work Package	Phase	Consensus Hrs	Precedence task(s)	Responsible person
1.1.1	Authentication	Frontend	5	2.2.1 Firebase Auth Integration	A.P
1.1.2	Scam Report Feed	Frontend	7	1.1.1 Authentication; 2.3.3 API Services	T.P
1.1.3	AI-Powered Search UI	Frontend	12	2.3.1 Gemini LLM Integration; 2.3.2 RAG Retrieval Logic	A.P/B.S
1.1.4	Report Submission Form	Frontend	8	1.1.1 Authentication; 2.2.2 Supabase Storage Integration; 2.3.3 API Services	T.P
1.1.5	Announcements Feature	Frontend / Admin	7	2.2.1 Firebase Auth Integration; 2.3.3 API Services	S.P
1.1.6	Data Privacy & ToS	Frontend	4	1.1.1 Authentication; 1.1.5 Report Submission Form	B.S
1.1.7	Support & Settings	Frontend	5	1.1.1 Authentication	B.S
1.2.1	Pending Review Queue	Admin	7	2.2.1 Firebase Auth Integration; 2.3.3 API Services	Y.R
1.2.2	Admin Action UI	Admin	8	1.2.1 Pending Review Queue	Y.R
2.1.1	PostgreSQL + pgvector	Infrastructure	7	None	A.P/T.P
2.1.2	Prisma ORM Implementation	Infrastructure	6	2.1.1 PostgreSQL +	A.P

WBS ID	Work Package	Phase	Consensus Hrs	Precedence task(s)	Responsible person
				pgvector	
2.2.1	Firebase Auth Integration	Auth	7	2.1.2 Prisma ORM Implementation	A.P
2.2.2	Supabase Storage Integration	Storage	7	2.2.1 Firebase Auth Integration	A.P/T.P
2.3.1	Gemini LLM Integration	AI	8	2.1.1 PostgreSQL + pgvector	A.P/B.S
2.3.2	RAG Retrieval Logic	AI	12	2.3.1 Gemini LLM Integration; 2.1.1 PostgreSQL + pgvector	A.P
2.3.3	API Services	Backend	12	2.1.2 Prisma ORM Implementation; 2.2.1 Firebase Auth Integration	ALL MEMBERS
3.1	Unit & Integration Testing	Testing	6	All development work packages	ALL MEMBERS
3.2	User Acceptance Testing	Testing	6	3.1 Unit & Integration Testing	ALL MEMBERS
3.3	CI/CD Pipeline & Deployment	Deployment	6	3.1 Unit & Integration Testing	T.P/S.P
4.1	Weekly Team Sync	PM	LOE		ALL MEMBERS
4.2	Progress Monitoring & Reporting	PM	LOE		ALL MEMBERS
	TOTAL		140		

Features Description

1.1.1 Authentication

Implements the full user authentication flow, covering the login and registration interfaces and secure session management. This package is a prerequisite for all personalised features.

1.1.2 Scam Report Feed

Merged from the former 1.1.2 Home Page / Public Feed and 1.1.3 Personal Feed packages. Both views render the same underlying list component; the differences are conditional columns, a status badge, and an auth guard that switches between the public and personal modes. Live statistics are displayed in the public view header.

1.1.3 AI-Powered Search UI

Presents users with a natural-language semantic search interface backed by the RAG engine, displaying ranked results that match query intent rather than exact keywords. Backend latency handling is scoped to 2.3.2.

1.1.4 Report Submission Form

Enables users to submit a new scam report through a validated form with evidence file upload. Consent acknowledgement is embedded in the flow. Note: the privacy policy text itself is scoped to 1.1.7.

1.1.5 Announcements Feature

Merged from the former 1.1.6 Announcement Page and 1.2.3 Announcement Management packages. Both access levels share the same data model and API calls. General users see a read-only list with category filtering; administrators see the same list with an additional form layer for creating, editing, and publishing announcements.

1.1.6 Data Privacy & ToS

Provides dedicated static pages for the data privacy policy and terms of service, and wires consent checkboxes into the registration and report submission flows.

1.1.7 Support & Settings

Delivers an onboarding tutorial for new users and an account settings panel where preferences and account details can be updated.

1.2.1 Pending Review Queue

Gives administrators a consolidated view of all reports awaiting review, with priority flagging and sort controls to help triage high-volume queues efficiently.

1.2.2 Admin Action UI

Renamed from 'Verification Controls' to avoid naming confusion with the fraud scoring logic in the AI layer. Provides the approve and reject action handlers with remark input, and a simple status-flag escalation control (not a separate workflow screen).

2.1.1 PostgreSQL + pgvector

Designs and provisions the core database schema for reports, announcements, users, and status history. The pgvector extension is configured on a managed Postgres instance to support similarity search for the RAG engine.

2.1.2 Prisma ORM Implementation

Wraps the PostgreSQL database with Prisma ORM, defining typed data models, managing schema migrations, and establishing the query layer consumed by all backend services.

2.2.1 Firebase Auth Integration

Integrates Firebase Authentication as the identity provider, supporting OAuth social logins, handling JWT token issuance and validation, and enforcing binary role-based access control (user / admin) across all protected routes.

2.2.2 Supabase Storage Integration

Sets up the evidence file upload pipeline using Supabase Storage, configures CDN delivery for efficient asset access, and enforces bucket policies to prevent unauthorised access.

2.3.1 Gemini LLM Integration

Connects the platform to Google Gemini as the underlying LLM, including crafting the initial prompt specification and API abstraction layer. Ongoing prompt tuning beyond the initial spec is treated as maintenance and is out of scope for this package.

2.3.2 RAG Retrieval Logic

Implements the retrieval-augmented generation pipeline for the bounded dataset of verified reports. Text chunks are embedded into vector representations, stored in pgvector, and retrieved by cosine similarity when a user submits a semantic query. Backend latency handling (e.g., streaming) is scoped here, not in the UI package.

2.3.3 API Services

Merged from the former 2.3.4 CRUD API Services and 2.4.1 Announcement API packages. Announcements are a core entity and share the same middleware, validation, and authentication patterns as report endpoints, making a separate package unnecessary. Exposes the full Create, Read, Update, Delete surface for all entities including reports, users, statuses, and announcements.

3.1 Unit & Integration Testing

Covers unit tests for all major frontend components and backend services, plus integration tests verifying that API endpoints interact correctly with the database and authentication layers.

3.2 User Acceptance Testing

Validates the platform against real user scenarios — including announcement browsing, report submission, scam feed browsing, and personal feed flows — and collects structured feedback to drive final issue resolution.

3.3 CI/CD Pipeline & Deployment

Establishes the automated build-and-deploy pipeline, validates functionality in a staging environment, and executes the production release with monitoring in place.

4.1 Weekly Team Sync

Ongoing weekly standups and sprint retrospectives to surface blockers and maintain team alignment throughout the project lifecycle. A Level-of-Effort (LOE) activity is excluded from the Delphi estimation.

4.2 Progress Monitoring & Reporting

Continuous maintenance of the project burndown chart, status reporting to the course instructor, and updates to the WBS and schedule as the project evolves. A Level-of-Effort (LOE) activity is excluded from the Delphi estimation.

6. Assumption List

6.1 Team and Availability

ID	Assumption
A-01	All five team members are available for the full duration of the project with no extended absences planned.
A-02	Each team member is expected to contribute approximately 10-15 productive hours per week toward development tasks.
A-03	No dedicated QA engineer is available; testing responsibilities are shared among all members.
A-04	External blockers (e.g., university exams, personal commitments) will be communicated at least one sprint in advance.
A-05	Everyone agreed to do feature-based work.

6.2 Technology and Infrastructure

ID	Assumption
A-06	The project will use Flutter as Frontend, Elysia.js as Backend and ORM as prisma.
A-07	Firebase Authentication is the sole identity provider; no custom auth server will be built.
A-08	Supabase Storage is pre-provisioned and accessible; bucket policies are configured before the integration sprint begins.
A-09	A managed PostgreSQL instance with the pgvector extension enabled is available before backend development starts.
A-10	Google Gemini API access (API key and quota) is granted to the team before the AI integration sprint begins.
A-11	All third-party APIs (Firebase, Supabase, Gemini) are stable, documented, and will not undergo breaking changes during the project.
A-12	The CI/CD pipeline (with github action) will target a single staging and one production environment; multi-region deployment is out of scope.

6.3 Scope and Definition of Done

ID	Assumption
A-13	A work package is considered done when it passes peer code review, has associated unit/integration tests, and is merged to the main branch.
A-14	Hour estimates cover active development time only; waiting time (e.g., API approval, PR review cycles) is not included.
A-15	UI/UX design assets and wireframes will be agreed upon before frontend implementation begins; design iteration time is not included in estimates.
A-16	Responsive design is required for tablet viewports and mobile
A-17	Localisation (i18n) and multi-language support are out of scope for this release.
A-18	The platform will support Thai and English content in the database, but the UI will be in English only.

ID	Assumption
A-19	The RAG engine operates over a bounded dataset of verified reports only; real-time external scam databases are out of scope.
A-20	Role-based access control is binary (user / admin); fine-grained permissions or multiple admin roles are out of scope.

6.4 Estimation Methodology

ID	Assumption
A-21	Estimates represent calendar hours of focused development effort, not elapsed wall-clock time.
A-22	The arithmetic mean of all five estimators' values is used as the baseline for consensus discussion.
A-23	Level-of-Effort (LOE) activities (Weekly Team Sync, Progress Monitoring) are excluded from the Delphi point total and managed separately.
A-24	Estimates do not include time for project management overhead, stakeholder meetings, or instructor check-ins.
A-25	A 10-15% contingency buffer may be applied at the project-plan level but is not embedded in individual WBS estimates.

7. Convergence Summary

7.1 Number of Rounds

The Wideband Delphi estimation process required two rounds before the team reached acceptable convergence. A third round was discussed but deemed unnecessary given the degree of alignment achieved after Round 2.

7.2 Round-by-Round Totals

Metric	Round 1	Round 2	Change
Team total — A.P.	129 hrs	128 hrs	-1
Team total — T.P.	135 hrs	130 hrs	-5
Team total — S.P.	133 hrs	131 hrs	-2
Team total — B.S.	126 hrs	140 hrs	+14
Team total — Y.R.	120 hrs	131 hrs	+11
Grand mean	128.6 hrs	132.2 hrs	+3.6
Range (max - min)	15 hrs	12 hrs	-3 (20% reduction)
Std. deviation (approx.)	~5.31 hrs	~4.15 hrs	22% reduction

7.3 Final summary

We start by having everyone look at a feature on their own. Each person writes down how many hours they think it will take, based on their own experience and how fast they work. This gives us a starting point based on everyone's skills.

Next, we come together as a team to share our numbers. We talk about why one person might think a task is quick, while another person thinks it will take longer. This is the time to point out things we might have missed, like difficult bugs, database setup, or extra testing.

After we have finished talking and everyone understands the work better, we do a second round of estimation. This time, we use everything we learned from the discussion to pick a more accurate number. This helps the whole team agree on a final timeline that we can all stick to.
